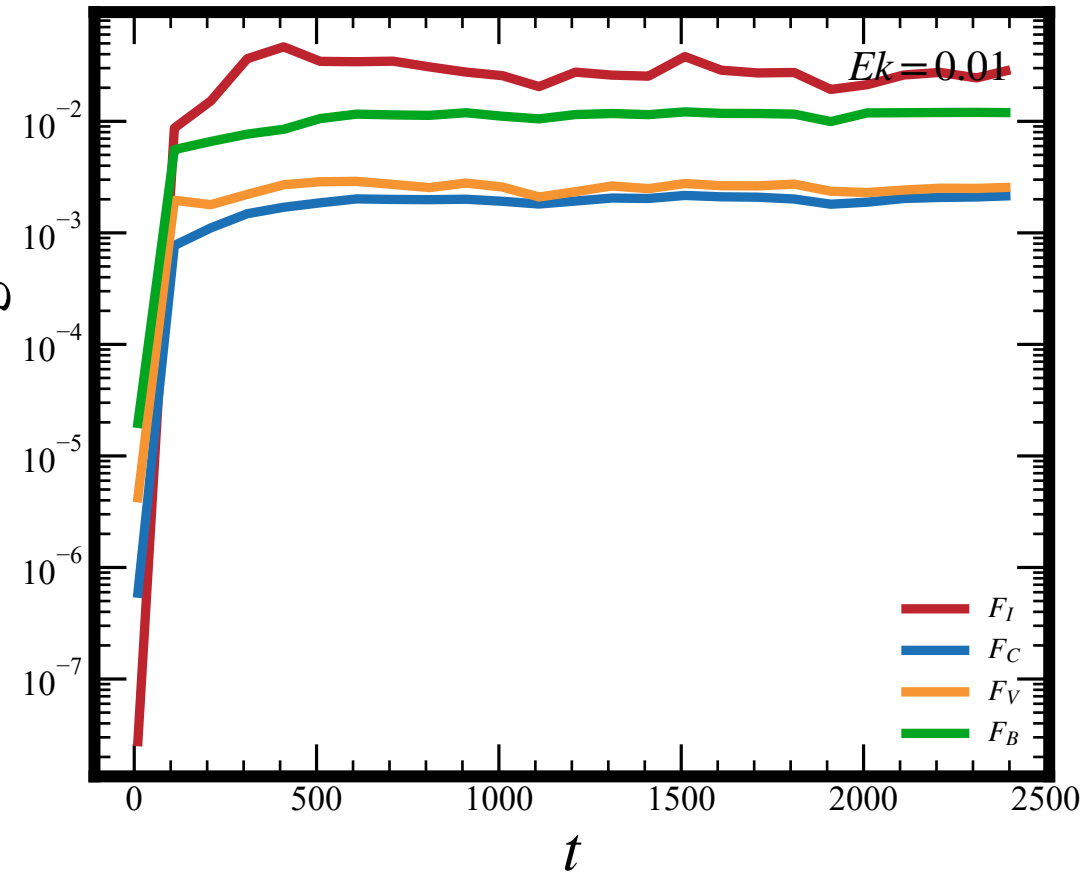
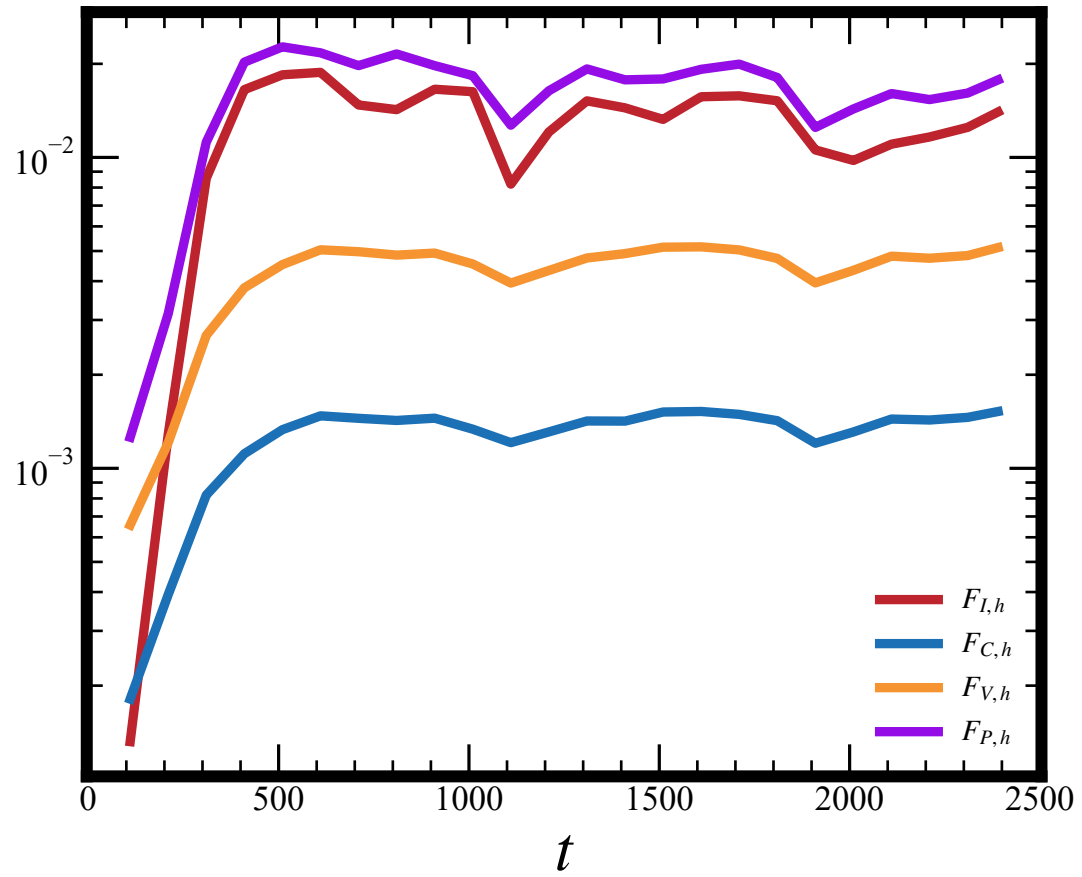


bulk force magnitude



horizontal force at $z \simeq 0.048$



horizontal force ratio

